



Strongly Connected Components and Topological Sort.

1 Topological Sort

Definition 1. (connectivity)

1. Let $G = (V, E)$ be a non-directed graph. A **connected component** of G is a subset $U \subseteq V$ of maximal size in which there exists a path between every two vertices.
2. A non-directed graph G is said to be a **connected** graph if it only has one connected component.
3. Let $G = (V, E)$ be a directed graph. A **strongly connected component** of G is a subset $U \subseteq V$ of maximal size in which for any pair of vertices $u, v \in U$ there exist both directed path from u to v and a directed path from v to u .

1.1 Depth First Search (DFS)

As its name implies, depth-first search searches "deeper" in the graph whenever possible. Depth-first search explores edges out of the most recently discovered vertex v that still has unexplored edges leaving it. Once all of v 's edges have been explored, the search "backtracks" to explore edges leaving the vertex from which v was discovered. This process continues until all vertices that are reachable from the original source vertex have been discovered. If any undiscovered vertices remain, then depth-first search selects one of them as a new source, repeating the search from that source. The algorithm repeats this entire process until it has discovered every vertex.

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1 DFS(  $G$  ):
2   for  $v \in V$  do
3      $v.\text{visited} \leftarrow \text{False}$ 
4   end
5    $\text{time} \leftarrow 1$ 
6   for  $v \in V$  do
7     if not  $v.\text{visited}$  then
8        $\pi(v) \leftarrow \text{null}$ 
9       Explore(  $G, v$  )
10    end
11  end

1 Explore( $G, v$ ):
2   Previsit( $v$ )
3   for  $(v, u) \in E$  do
4     if not  $u.\text{visited}$  then
5        $\pi(u) \leftarrow v$ 
6       Explore(  $G, u$  )
7     end
8   end
9   Postvisit( $v$ )

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1 Previsit( $v$ ):
2    $\text{pre}(v) \leftarrow \text{time}$ 
3    $\text{time} \leftarrow \text{time} + 1$ 

1 Postvisit( $v$ ):
2    $\text{post}(v) \leftarrow \text{time}$ 
3    $\text{time} \leftarrow \text{time} + 1$ 

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Properties of depth-first search. Depth-first search yields valuable information about the structure of a graph. Perhaps the most basic property of depth-first search is that the predecessor subgraph G_π does indeed form a forest of trees since the structure of the depth-first trees exactly mirrors the structure of recursive calls of explore-function. That is, $u = \pi(v)$ if and only if $\text{explore}(G, v)$ was called during a search of u 's adjacency list.

Additionally, vertex v is a descendant of vertex u in the depth-first forest if and only if v is discovered during the time in which $\text{pre}(u)$ has already been set but $\text{post}(u)$ hasn't¹. Another important property of depth-first search is that discovery and finish times have a parenthesis structure. If the explore procedure were to print a left parenthesis " u " when it discovers vertex u and to print a right parenthesis " u " when it finishes u , then the printed expression would be well-formed in the sense that the parentheses are properly nested.

The following theorem provides another way to characterize the parenthesis structure.

Theorem 1 (Parenthesis theorem). *In any depth-first search of a (directed or undirected) graph $G = (V, E)$, for any two vertices u and v , exactly one of the following three conditions holds:*

1. *the intervals $[\text{pre}(u), \text{post}(u)]$ and $[\text{pre}(v), \text{post}(v)]$ are entirely disjoint, and neither u nor v is a descendant of the other in the depth-first forest.*
2. *the interval $[\text{pre}(u), \text{post}(u)]$ is contained entirely within the interval $[\text{pre}(v), \text{post}(v)]$, and u is a descendant of v in a depth-first tree, or*

¹In terms of the presentation from last week, when u 's color is gray